

# Ocular Complications after Iodine Brachytherapy for Large Uveal Melanomas

Ilkka Puusaari, MD,<sup>1</sup> Jorma Heikkonen, PhD,<sup>2</sup> Tero Kivelä, MD<sup>1</sup>

**Purpose:** To evaluate anterior and posterior segment complications and their management after iodine 125 plaque brachytherapy (IBT) for large uveal melanoma.

**Design:** Retrospective nonrandomized interventional study.

**Participants:** Ninety-six patients with a large uveal melanoma according to the Collaborative Ocular Melanoma Study criteria.

**Methods:** The patients underwent primary IBT (median dose to tumor apex, 87 Gy). The median tumor height and diameter were 10.7 mm (range, 4.5–16.8) and 16.5 mm (range, 7.3–25.0), respectively, and the median follow-up time was 3.5 years (range, 0.3–10.4). Cumulative incidence analysis and competing risks regression were used to analyze the time to individual complications and to identify risk factors. Death and secondary enucleation were analyzed as competing risks.

**Main Outcome Measures:** Cataract, iris neovascularization, glaucoma, maculopathy, optic neuropathy, vitreous hemorrhage, and persistent exudative retinal detachment (RD).

**Results:** The 5-year cumulative incidences of cataract, iris neovascularization, and glaucoma were 69% (95% confidence interval [CI], 57%–78%), 62% (95% CI, 50%–71%), and 60% (95% CI, 48%–70%), respectively. Posterior segment complications were less common. The 5-year incidences of maculopathy and optic neuropathy were 52% (95% CI, 35%–65%) and 46% (95% CI, 30%–61%), and those of vitreous hemorrhage and persistent RD were 36% (95% CI, 23%–48%) and 25% (95% CI, 15%–36%), respectively. More than 80% of complications were diagnosed within 3 years. Cataract was the earliest complication to appear. Except for cataract, the cumulative incidence of dying without developing a particular complication was 0.24 to 0.62 times that of first developing the complication. Increasing tumor height, which correlates to increasing doses to adjacent tissues, was associated with time to cataract ( $P = 0.017$ ), iris neovascularization ( $P = 0.087$ ), and RD ( $P = 0.046$ ). Maculopathy and optic neuropathy were associated primarily with distance to the fovea ( $P = 0.015$ ) and optic disc ( $P = 0.015$ ), respectively. Of 57 patients with cataract, 47% underwent cataract extraction, and 12% of 51 patients with glaucoma were treated with cyclophotocoagulation. The prevalences of cataract, elevated intraocular pressure, and RD were 43%, 39%, and 13%, respectively, at 5 years.

**Conclusions:** The frequency with which ocular complications develop after IBT is notably influenced by competing risks. Cumulative incidence and prevalence analysis provide realistic estimates for patient counseling. *Ophthalmology* 2004;111:1768–1777 © 2004 by the American Academy of Ophthalmology.

A patient with a large choroidal and ciliary body melanoma faces blindness and loss of the affected eye both by the tumor and by the treatment. The most common treatments currently available for a large uveal melanoma are enucle-

ation, iodine brachytherapy (IBT), and proton beam therapy.<sup>1–3</sup> The effectiveness of enucleation and IBT seem comparable in terms of survival for medium-sized to large uveal melanoma,<sup>4–7</sup> but although enucleation carries a smaller risk of subsequent complications and local recurrence, brachytherapy offers a fair chance of saving the eye and some chance of preserving useful vision at least short-term.<sup>7,8</sup> For every 6 patients with a large melanoma managed with IBT, 1 avoids blindness of the tumor eye for  $\geq 2$  years.<sup>7</sup>

Despite unfavorable odds of preserving vision long-term, many patients now prefer IBT to primary enucleation.<sup>7,9</sup> In making the decision, the patient must balance the benefit of saving the eye and some vision against the risk of recurrence and the inconvenience of complications and their treatment. To counsel the patient, the ophthalmologist needs realistic data on what complications likely develop, and how they will affect visual function, appearance, and eye retention.

Originally received: September 26, 2003.

Accepted: March 5, 2004.

Manuscript no. 230635.

<sup>1</sup> Ocular Oncology Service, Department of Ophthalmology, Helsinki University Central Hospital, Helsinki, Finland.

<sup>2</sup> Department of Oncology, Helsinki University Central Hospital, Helsinki, Finland.

Supported by The Finnish Eye Foundation, The Eye and Tissue Bank Foundation, Sigrid Jusélius Foundation, Ida Montin Foundation, Finnish Medical Foundation, and The Helsinki University Central Hospital Research Fund (grant no.: TYH3203), Helsinki, Finland.

The authors have neither conflicts of interest nor proprietary interests in the technique or equipment described.

Correspondence to Ilkka Puusaari, MD, Department of Ophthalmology, Helsinki University Central Hospital, PL 220, Haartmaninkatu 4 C, FIN-00029 HUS, Helsinki, Finland. E-mail: ilkka.puusaari@hus.fi.